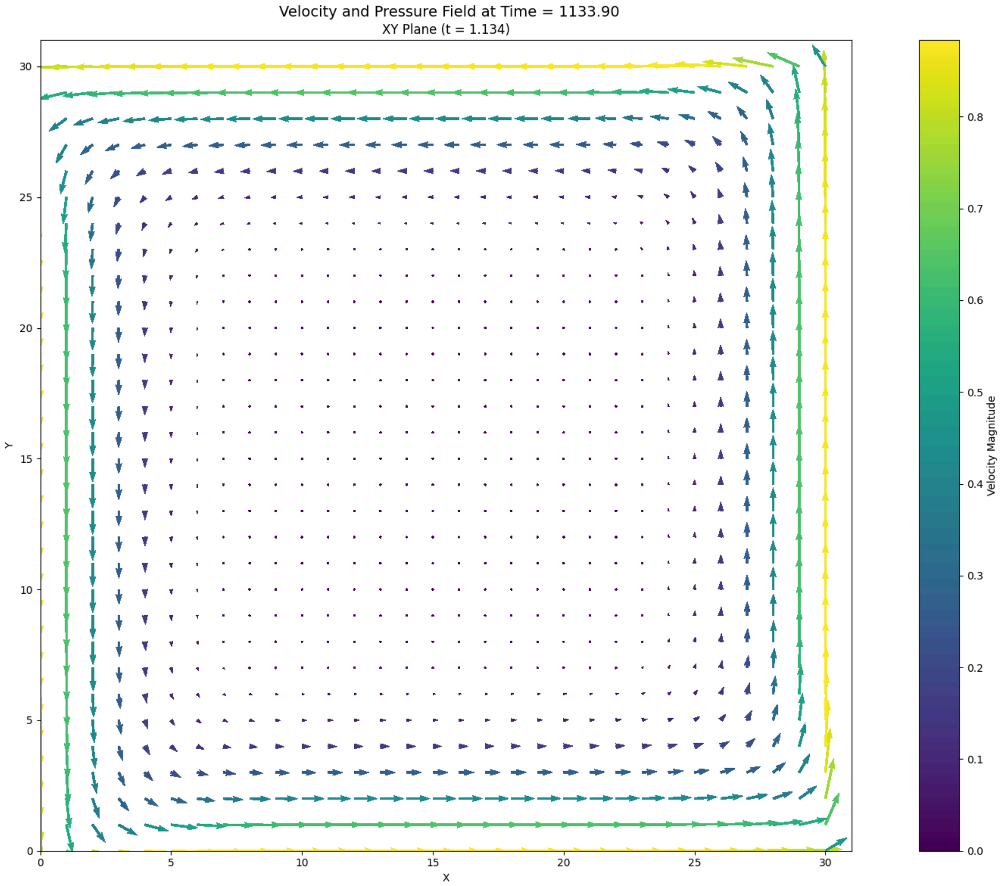
Experimental Report

Case: Lid Cavity with 4 Moving Walls

Description of Flow: This experiment studies a two-dimensional cavity flow in which all four walls move at prescribed speeds, setting the fluid into a strong recirculating motion. The moving boundaries inject momentum into the fluid and generate a nonlinear vortex structure with both shear-driven and pressure-driven effects. The objective is to identify the governing streamwise momentum equation directly from the evolving velocity and pressure fields.

> Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

```
u_t =- 1.00000000e+00*uu_x - 1.00000000e+00*vu_y + 1.00000000e-02*u_xx +
1.00000000e-02*u_yy - 1.00000000e-03*p_x
```

Learned equation

```
u_t =- 1.39217942e+00*uu_x - 9.75843704e-01*vu_y + 2.41619947e-02*u_xx +
1.02176557e-02*u_yy - 9.61020893e-04*p_x
```

Term	Reference coefficient	Learned coefficient	Absolute error
uu_x	−1.00000000e+00	−1.39217942e+00	3.92179419e−01
vu_y	−1.00000000e+00	−9.75843704e−01	2.41562962e−02
u_xx	+1.00000000e−02	+2.41619947e−02	1.41619947e−02
u_yy	+1.00000000e−02	+1.02176557e−02	2.17655709e−04
p_x	−1.00000000e−03	−9.61020893e−04	3.89791068e−05

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	rho	1.00000000e+03	kg m^{-3}
Dynamic viscosity	viscosity	1.00000000e+01	Pa s
Kinematic viscosity	nu	1.00000000e−02	m^2 s^{-1}
Characteristic velocity	u_max	1.00000000e+00	m s^{-1}
Characteristic length	D	1.00000000e+00	m
Reynolds number	Re	1.00000000e+02	dimensionless
Body force	G	9.81000000e+00	m s^{-2}
Grid points in x	Nx	3.10000000e+01	cells
Grid points in y	Ny	3.10000000e+01	cells
Grid spacing in x	dx	3.22580645e−02	m
Grid spacing in y	dy	3.22580645e−02	m
Time step	dt	1.00000000e−04	s
Numerical threshold	epsilon	7.00000000e−11	-

3. Numerical Discretisation

Quantity	Symbol	Value	Unit
Parameters listed above are the values used by the selected solver.			

4. Data and Pre-processing

Quantity	Value
Loaded field shape (t, x, y)	(9851, 30, 31)
Regression field shape (t, x, y)	(7451, 22, 27)
Regression samples	4,425,894

Quantity	Value
Dictionary terms	5
First saved time index	150
Temporal crop at each boundary	1200 layers
x-boundary crop	4 points
y-boundary crop	2 points

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	9.42668455e-04
L0 penalty	1.00000000e-06
Training RMSE	1.99802752e-02
Validation RMSE	2.52969478e-02